



ParceLops Functional & Non-Functional Requirements

1. Functional Requirements

1.1 Authentication & User Management

FR-001 User Registration

The system shall allow:

- Parcel company admins to create company accounts
- Workers to be invited by admins
- Customers to register using phone number/email

FR-002 Login

The system shall support:

- Email/password login
- Phone number login
- OTP authentication
- Multi-factor authentication (MFA) for admins/workers

FR-003 Role-Based Access Control (RBAC)

The system shall support these roles:

- Super Admin
- Company Admin
- Branch Manager
- Worker/Counter Staff
- Customer

Each role shall only access authorized resources.

FR-004 Session Management

The system shall:

- Automatically expire inactive sessions
- Support refresh tokens
- Allow logout from all devices

1.2 Parcel Intake & Registration

FR-005 Barcode/QR Generation

The system shall:

- Generate unique barcode/QR codes for parcels
- Associate codes with parcel records

FR-006 Parcel Scanning

Workers shall be able to:

- Scan incoming parcels
- Scan outgoing parcels
- Scan collected parcels

FR-007 Automatic Parcel Registration

Upon scanning:

- Parcel data shall automatically register in the system
- Timestamp and worker ID shall be logged

FR-008 Parcel Metadata

The system shall store:

- Sender details
- Receiver details
- Parcel type
- Tracking number
- Weight
- Branch location
- Status

1.3 Parcel Tracking

FR-009 Real-Time Tracking

Customers shall:

- View parcel status in real time
- Track parcel movement history

FR-010 Status Workflow

The system shall support statuses:

- Pending
- In Transit
- Ready for Pickup
- Collected
- Delivered
- Returned
- Lost

FR-011 Tracking History

The system shall maintain:

- Full audit trail
- Scan history
- Location history
- Worker actions

1.4 Pickup Scheduling System

FR-012 Pickup Slot Booking

Customers shall:

- View available pickup slots
- Reserve pickup times

FR-013 Queue Management

The system shall:

- Limit bookings per slot
- Prevent overcrowding

FR-014 Slot Prediction

The system should:

- Predict high-traffic periods
- Recommend optimal pickup times

FR-015 Rescheduling

Customers shall:

- Reschedule pickup appointments
- Cancel bookings

1.5 Notifications

FR-016 SMS Notifications

The system shall send SMS when:

- Parcel arrives
- Parcel status changes
- Pickup slot is near
- Delivery is completed

FR-017 Push Notifications

The mobile app shall support push notifications.

FR-018 Notification Preferences

Users shall:

- Enable/disable notifications
- Choose SMS or push

1.6 Delivery & Outbound Flow

FR-019 Delivery Request

Customers shall:

- Request home delivery
- Select delivery address

FR-020 Proof of Delivery (POD)

Workers shall capture:

- Customer signature
- Delivery photo

FR-021 Delivery Confirmation

The system shall:

- Confirm successful delivery
- Record delivery timestamps

1.7 Payment System

FR-022 Payment Integration

The system shall integrate with:

- Mobile Money (MoMo)
- Paystack
- Hubtel

FR-023 Payment Processing

Customers shall:

- Pay delivery fees in-app
- Receive receipts

FR-024 Refund Workflow

The system shall:

- Support refunds
- Require authorization for refunds

1.8 Worker Dashboard

FR-025 Dashboard Overview

Workers shall view:

- Daily parcel intake
- Pending pickups
- Delivery requests

FR-026 Parcel Search

Workers shall search by:

- Tracking number
- Phone number
- Barcode

FR-027 Scan Logs

Workers shall view scan activity history.

1.9 Admin Dashboard

FR-028 Company Analytics

Admins shall view:

- Parcel volumes
- Queue reduction metrics
- Delivery success rates
- Revenue metrics

FR-029 Branch Management

Admins shall:

- Create branches
- Assign workers
- Configure branch settings

FR-030 Reporting

The system shall generate:

- Daily reports
- Monthly reports
- Operational analytics

1.10 Multi-Tenant Architecture

FR-031 Tenant Isolation

Each company's data shall be isolated using org_id scoping.

FR-032 Tenant Configuration

Each company shall:

- Have branding settings
- Manage users independently

1.11 Offline Support

FR-033 Offline Scanning

Workers shall scan parcels without internet access.

FR-034 Sync Engine

Offline data shall synchronize automatically once connected.

1.12 Audit & Compliance

FR-035 Audit Logs

The system shall log:

- Login attempts
- Parcel scans
- Payment events
- Data exports

FR-036 Data Retention

The system shall:

- Retain records based on policy
- Support data deletion requests

2. Non-Functional Requirements

2.1 Performance Requirements

NFR-001 Response Time

- API responses shall complete within 2 seconds under normal load.

NFR-002 Scan Speed

- Barcode scans shall process within 1 second.

NFR-003 Concurrent Users

- The system shall support at least 5,000 concurrent users initially.

NFR-004 Scalability

- The architecture shall scale horizontally.

2.2 Availability & Reliability

NFR-005 System Uptime

- The platform shall maintain 99.9% uptime.

NFR-006 Backup & Recovery

- Daily automated backups shall be performed.

NFR-007 Disaster Recovery

- Recovery Time Objective (RTO): < 4 hours
- Recovery Point Objective (RPO): < 15 minutes

2.3 Security Requirements

NFR-008 Encryption

The system shall use:

- TLS 1.3 for data in transit
- AES-256 for data at rest

NFR-009 Password Security

Passwords shall:

- Be hashed using bcrypt or Argon2

- Never be stored in plaintext

NFR-010 Multi-Factor Authentication

MFA shall be mandatory for:

- Admin accounts
- Worker accounts

NFR-011 Access Control

The system shall enforce:

- RBAC
- Least privilege access

NFR-012 Auditability

Every critical action shall be traceable.

NFR-013 Secure APIs

APIs shall support:

- Rate limiting
- JWT authentication
- Token rotation

2.4 Usability Requirements

NFR-014 Ease of Use

Workers with minimal technical experience shall operate the app after basic training.

NFR-015 Mobile Optimization

The platform shall work efficiently on:

- Low-end Android devices
- Small screens

NFR-016 Accessibility

The UI shall support:

- Large touch targets
- Clear visual feedback

2.5 Network & Offline Requirements

NFR-017 Offline-First Capability

Core operations shall continue during internet outages.

NFR-018 Low Bandwidth Optimization

The system shall minimize data usage.

2.6 Maintainability

NFR-019 Modular Architecture

The system shall use modular services/components.

NFR-020 API Documentation

All APIs shall be documented.

NFR-021 Logging

The platform shall maintain centralized logs.

2.7 Compatibility Requirements

NFR-022 Cross-Platform Support

The client app shall support:

- Android

- iOS
- Web browsers

NFR-023 Browser Support

The web platform shall support:

- Chrome
- Edge
- Safari
- Firefox

2.8 Compliance Requirements

NFR-024 Ghana Data Protection Compliance

The platform shall comply with:

- Ghana Data Protection Act 2012

NFR-025 Data Privacy

Analytics data shall be anonymized before sharing.

NFR-026 Regulatory Readiness

The system should support future:

- GDPR compliance
- PCSRC compliance requirements

2.9 Monitoring & Observability

NFR-027 Monitoring

The platform shall monitor:

- API failures
- Login anomalies
- Scan anomalies

NFR-028 Alerting

Critical failures shall trigger real-time alerts.

NFR-029 Metrics Collection

The system shall collect:

- Parcel scans/day
- Queue reduction metrics
- Active companies
- Churn rate

2.10 Business Requirements

NFR-030 Multi-Tenant SaaS

The platform shall support multiple parcel companies on one infrastructure.

NFR-031 White Label Support

The platform should allow:

- Company logos
- Custom branding
- Custom domains

NFR-032 Localization

The system should support:

- Ghanaian phone formats
- GHS currency
- MoMo workflows

3. Future Functional Requirements (Phase 2+)

FR-FUTURE-001 AI Queue Prediction

Predict branch congestion using historical data.

FR-FUTURE-002 Route Optimization

Optimize delivery routes automatically.

FR-FUTURE-003 Enterprise APIs

Expose APIs for large courier integrations.

FR-FUTURE-004 Analytics Marketplace

Provide anonymized logistics intelligence dashboards.

FR-FUTURE-005 Insurance Integration

Offer parcel insurance products through partners.

THANK YOU !